



Clinical short communication

Early deterioration of CIDP following transition from IVIG to FcRn inhibitor treatment

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ABSTRACT

Chronic inflammatory demyelinating polyneuropathy (CIDP) is an immune mediated demyelinating neuropathy that can lead to secondary axonal degeneration and irreversible weakness and disability. Early effective treatment is therefore necessary to minimize the degree of axonal degeneration. Prior to 2024 the only FDA approved therapy for CIDP was intravenous immunoglobulin (IVIg). In 2024, efgartigimod (Vyvgart-Hytrulo), a FcRn inhibiting therapy (FIT) was approved for treatment of CIDP based on the phase II Adhere study. In the controlled setting of the phase II clinical study patients who were stable on IVIg were taken off treatment to ensure that their disease was active, and patients who worsened were then treated with efgartigimod. The responders were then randomized (in phase B) to either placebo or continued efgartigimod treatment. In the real world setting it is not feasible to stop IVIg and let patients worsen before starting a FIT, thus the transition from IVIg to efgartigimod in a real world setting was not studied in the pivotal trial. We have treated nine patients with FIT in our practice and report findings of four of those patients who had severe relapse of CIDP after treatment. Five of the other patients neither improved nor declined with FIT. This raises questions about the issues related to transitioning patients from IVIG to efgartigimod.

1. Introduction

Chronic inflammatory demyelinating polyneuropathy (CIDP) is an immune mediated demyelinating neuropathy that can lead to secondary axonal degeneration and irreversible weakness and disability [1]. Early effective treatment is therefore necessary to minimize the degree of axonal degeneration. Prior to 2024 the only FDA approved therapy for CIDP was intravenous immunoglobulin (IVIg) [2]. In 2024, efgartigimod (Vyvgart-Hytrulo), a FcRn inhibiting therapy (FIT) was approved for treatment of CIDP based on the phase II Adhere study [3]. In the controlled setting of the phase II clinical study patients who were stable on IVIg were taken off treatment to ensure that their disease was active, and patients who worsened were then treated with efgartigimod. The responders were then randomized (in phase B) to either placebo or continued efgartigimod treatment. In the real world setting it is not feasible to stop IVIg and let patients worsen before starting a FIT, thus the transition from IVIg to efgartigimod in a real world setting was not studied in the pivotal trial. We have treated nine patients with FIT in our practice and report findings of four of those patients who had severe relapse of CIDP after treatment. Five of the other patients neither

improved nor declined with FIT. This raises questions about the issues related to transitioning patients from IVIG to efgartigimod.

2. Materials and methods

The case series was collected as part of our clinical practices at the neuroscience Institute. The information is related to care that patient has received as part of their standard of care (See Table 1).

2.1. Results/case series

Case 1. A 68-year-old woman with a three-year history of typical CIDP (satisfying EFNS/PNS clinical and electrophysiological criteria) had been stable on IVIg therapy for the last one year.

At initial presentation three years ago, her INCAT disability score was 5 and she needed a walker for ambulation and had severe difficulty using her arms and hands in daily activities.

Her initial treatment was IVIG which led to partial improvement (70 %) with an INCAT disability score of 2. Therefore, prednisone was

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added, and she had further improvement in strength over the next six months with her INCAT score improving to a 1. Her prednisone was gradually weaned off over another six months, and she remained stable on IVIG monotherapy at 1 g/kg/day every three weeks.

Given her continuing mild distal leg weakness with ankle dorsiflexion strength of 4/5 and INCAT score of 1 a decision was made to switch her to efgartigimod to look for further improvement.

She received her first injection 12 days after her last dose of IVIG. Two days after infusion she noticed an increase in sensory dysesthesias in her feet extending to the mid-calf level. The day after her second dose of efgartigimod (day 8 of initiation of therapy) she noticed increased weakness proximally and distally in her legs which progressed to the arms before her third weekly dose. After her third weekly dose she was no longer able to ambulate normally and required assistance for her ADLs. The leg weakness and imbalance led to a tibial fracture after a fall. She was admitted to the hospital and at this point her MRC scores in all four limbs were 2–3/5 and INCAT score was 7. She received 10 cycles of plasmapheresis and began to have some improvement but still required being discharged to a rehab facility where she was started back on IVIG. Eight weeks after the treatment with efgartigimod had been initiated she remained with an INCAT score of 5 due to the need to use a walker and difficulty performing routine activities with her hands.

Case 2. A 74-year-old man with distal variant of CIDP (satisfying EFNS/PNS clinical and electrophysiological criteria) had been on a stable dose of IVIG for the last seven years.

At initial presentation he had distal arm and leg weakness and numbness with INCAT disability score of 4.

He responded to IVIg and INCAT score improved to 1 over the initial six months of treatment and he had remained stable since. He had weakness in ankle dorsiflexion (MRC 4) but was able to walk independently and he had no residual symptoms in his arms.

He requested a switch to efgartigimod to look for further improvement and efgartigimod was started six weeks after his last IVIg dose. During this delay between therapies, he noticed increased numbness in his legs, but he remained able to walk independently. Three days after his first dose of efgartigimod he noticed weakness in his hands and proximal arms. After the second dose (8 days after efgartigimod initiation) his arms worsened, and he could not lift them over his shoulders (should abductors MRC 2), and he began to have worsening of his ankle dorsiflexors. Two days after his third dose (17 days after efgartigimod initiation) he required a walker to walk and could not independently care for himself due to his arm weakness (INCAT score of 6). Efgartigimod was stopped and he was placed urgently back on IVIg. Within 2 weeks of IVIg he started improving and at 6 weeks he has regained near baseline strength in his arms except for mild impairment of dexterity in his hands. He remains with residual leg weakness requiring a walker for ambulation (INCAT 4).

Case 3. A 58-year-old female with a 15-year-history of typical CIDP (satisfying EFNS clinical and electrophysiological criteria) had been stable on IVIg therapy for the last ten years.

At the onset of her disease 15 years ago she was hospitalized with profound quadriplegia and INCAT score of 8. She was treated with plasma exchange and improved. She then started IVIG. She improved on

IVIG and remained stable. Five years previous, she had participated in a research study for CIDP during which she had worsening of strength and withdrew and went back to IVIG. Her INCAT score remained stable at 2, on IVIG 1 g/kg/day every three weeks. When the IVIG dosing interval was tapered to more than 4 weeks she had worsening of weakness.

Due to fatigue from the IVIG infusions she chose to switch to efgartigimod. Efgartigimod was started 20 days after her last IVIG infusion. Within 3 days of her first dose, she had an increase in her sensory symptoms and worsening gait ataxia. After the second dose (day 9) she had a dramatic increase in her weakness to the point that she had proximal arm and leg weakness that limited her from doing her ADLs and she required a cane for ambulation (INCAT score of 4). FIT was stopped and she was placed back on IVIG and within two weeks she no longer needed the cane but still had limitations in arms and legs proximally and distally. Two weeks after the IVIG she had an increase in her weakness (INCAT of 7) that necessitated being admitted into the hospital for urgent plasmapheresis. This therapy is ongoing.

Case 4. A 47-year-old female with a five-year history of multifocal CIDP (satisfying EFNS clinical and electrophysiological criteria) was stable on IVIG for the last four years.

Her initial presentation five years ago was of progressive asymmetric upper extremity predominant weakness and numbness along with hyporeflexia. Her INCAT score was 5 and her initial treatment was with IVIG and within 3 months she had improvement of left hand strength. Prednisone was added and she had significant improvement in strength over the next 6 months. She developed adverse effects to prednisone and mycophenolate was started and prednisone was tapered off. She continued on a combination of IVIG and mycophenolate and has remained on that regimen for the last 4 years, with an INCAT of 2. She did have one period where, due to insurance reasons the IVIG interval increased beyond six weeks, and she noticed increased hand weakness.

Due to the persistent weakness in her hands despite long term IVIG therapy she was switched to efgartigimod. Within three days of the first dose, she had increased weakness and numbness in her right arm and both legs. After the second dose (9 days after initiation) she could no longer independently drive or walk and unfortunately, she had a fall leading to a fracture at the ankle. Her INCAT score was 6. She was taken off efgartigimod and restarted on IVIG. After two weeks she had not noticed improvement and prednisone was added and follow up was scheduled for four weeks but has not yet occurred.

3. Discussion

In summary, four CIDP patients who were transitioned from IVIG to FIT worsened precipitously. At the time of this report, the patients have not yet returned to baseline with reinstatement of IVIg therapy and in some cases the addition of prednisone. In CIDP patients who relapse, there is no guarantee that they will return to baseline and therefore a relapse is always of concern given that some patients may have worsening of permanent disability as was seen in previous studies in CIDP [4].

Our cases here highlight a potential risk of switching patients from IVIG to FIT. In all four cases the patients had definite CIDP, and recent evidence of disease activity as evidenced by wearing off symptoms

Table 1.

Clinical features including disability scores before and after FIT treatment and after rescue treatment

Pt	Age/ Sex	Disease Duration (months)	Disease Characteristic	Duration on IVIg (months)	INCAT at onset	INCAT on IVIg	INCAT after FIT	Rescue Treatment	Time on Rescue therapy	INCAT after rescue
1	68/F	36	Classic	12	5	2	7	Plex, IVIg, Pred	9 wks	3
2	74/M	84	Distal	84	4	1	6	IVIg	8 wks	3
3	58/F	180	Classic	120	8	2	4	Plex, IVIg, Pred	6 wks	5
4	47/F	60	Multifocal	48	5	2	6	IVIg, Pred	5 wks	4

M = male, F = female, Pt = Patient, IVIg = intravenous immunoglobulin, Pred = prednisone, plex = plasma exchange.

between IVIG infusions and/or worsening with interruption of therapy. In addition, worsening of patients when IVIg was stopped provided retrospective evidence of active disease. The rapid worsening that coincided with initiation of FIT suggests a withdrawal effect from the IVIG given that FIT decreases serum IgG levels by blocking its salvage mechanism in the endothelial cells. Another concern is that the decline seen in these patients is faster and more severe than the wearing off or relapse seen in patients where IVIg is tapered in dose or frequency. Therefore, in addition to rapid serum IgG reduction, this also raises the possibility of other humoral factors or downstream cytokines that may be affected with FIT in patients with active CIPD.

In contrast to what we have observed in this small number of patients with CIPD, we are unaware of any reports, nor have we seen in our practices, patients with myasthenia gravis (MG) worsening when transitioned from IVIg to FIT. This suggests that the mechanism of action of efgartigimod in CIPD may be different than that for MG, or it could be related to the differing immunopathogenesis of these diseases. Further study of CIPD patients who experience dramatic worsening with FIT initiation with more detailed analysis of serum biomarkers, especially Ig and neurofilament levels, and nerve conduction studies may be beneficial [5].

We recommend that once a decline is noticed after starting FIT, therapy should be discontinued, and rescue therapy should be instituted as early as possible. In general, some strategies that could be considered in switching real world CIPD patients from IVIg to efgartigimod, include bridging with corticosteroids until the positive effect of FCRn inhibition on the disease process can potentially negate the effects of IVIG withdrawal or continue IVIg treatment during the initiation of efgartigimod therapy. There is a critical need to study this further.

4. Conclusion

In conclusion the transition of patients with CIPD who are stable on IVIG therapy to FIT can lead to severe worsening of disease. Further research to understand the immunopathogenesis of this worsening, may shed light as to the appropriate strategy for initiating this switch or may identify unique patient and disease characteristics that can predict this

decline. Recognition of the potential for severe worsening during this switch is important to investigate strategies that may prevent severe relapse and irreversible disability.

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Todd Levine: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Suraj Muley:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation.

Declaration of competing interest

none.

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